DIABETES PREDICTION USING LOGISTIC REGRESSION

MINI PROJECT REPORT

Student Name	Prajwal Reddy K	**Date**	1/11/2026

1. Project Overview

This project builds a binary classification model using Logistic Regression to predict the diabetes outcome label (0 or 1) from medical measurement features in the Pima Indians Diabetes dataset. The notebook contains 768 records and 9 columns, including 8 input features and the Outcome target. This is an educational machine-learning exercise, not a medical diagnostic tool.

2. Methodology

The dataset was loaded and explored using Pandas, NumPy, Matplotlib and Seaborn. Zero values in Glucose, BloodPressure, SkinThickness, Insulin and BMI were treated as missing values. The data was separated into input features (X) and target (y), then split into 80% training data (614 rows) and 20% testing data (154 rows) using stratification. Missing values were filled with the median calculated from the training data, and StandardScaler was used for feature scaling. A class-weight-balanced Logistic Regression model was trained and evaluated on the test set.

3. Model Evaluation

Metric	Result	Meaning
Accuracy	73.38%	Overall test predictions correct
Precision (class 1)	60.32%	Share of positive predictions that were correct
Recall (class 1)	70.37%	Share of actual positive cases identified
F1-score (class 1)	64.96%	Balance of precision and recall
ROC-AUC	0.813	Ability to rank positive cases above negative cases

Confusion Matrix: [[75, 25], [16, 38]] — 75 true negatives, 25 false positives, 16 false negatives and 38 true positives. The notebook also compares balanced and normal Logistic Regression: the balanced model achieved higher class-1 recall (0.70 vs 0.50) and higher overall accuracy (0.73 vs 0.71) on this test split.

4. Key Findings and Conclusion

Among the model coefficients, Glucose had the largest positive coefficient (1.183), followed by BMI (0.710). This means these features had the strongest positive contributions to the model's predicted class-1 score in this fitted model; coefficients do not establish causation. The balanced Logistic Regression model achieved 73.38% accuracy and an ROC-AUC of 0.813 on the held-out test set. The results demonstrate a complete machine-learning workflow involving data cleaning, imputation, scaling, training and evaluation. The dataset is limited and represents a specific population, so these results should not be generalized as medical advice or used to make real-world diagnosis decisions.

Tools & Libraries: Python, Google Colab, Pandas, NumPy, Matplotlib, Seaborn and Scikit-learn.